PROJECT AND TEAM INFORMATION

Project Title

Emergency Rescue Simulator

Student / Team Information

Team Name: *Team #*	*Avengers*
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

During emergencies such as fires, road accidents, medical emergencies, floods, and building collapses, multiple incidents may occur simultaneously. Emergency response centers have limited teams, vehicles, equipment, and other resources. Therefore, responding to every incident efficiently requires proper prioritization, resource allocation, and route planning.
The main motivation of this project is to develop a simulation system that demonstrates how computational techniques can assist an emergency control center in managing multiple emergency situations. The system will receive emergency incidents, determine their severity, prioritize them, identify the appropriate available response team, and determine an efficient route from the team's current location to the emergency location.
The project is particularly important from a Computer Science perspective because emergency management involves several problems that can be represented using Data Structures and Algorithms and how Object-Oriented Programming concepts in C++ can be used to model emergency types, response teams, locations, resources, and system operations.

State of the Art / Current solution (1 pt)

In real-world emergency management, emergency response is generally coordinated through centralized control or dispatch centers. When an emergency call is received, an operator collects information about the incident, determines its urgency, identifies suitable available resources, and dispatches the required teams. Modern systems may use GPS, GIS-based mapping, communication networks, traffic information, and specialized dispatch software to support these decisions.
For example, ambulance, fire, and police dispatch systems maintain information about available units and their locations and may use digital maps and routing systems to determine suitable routes.
However, such real-world systems are complex, expensive, and depend on infrastructure such as live GPS data, communication networks, geographic databases, and real-time traffic information.

Project Goals and Milestones (2 pts)

The primary goal of the project is to develop an Emergency Rescue Simulator that models the operation of an emergency control center and demonstrates the practical application of Data Structures, Algorithms, and Object-Oriented Programming.

The system will:

Register and maintain multiple emergency incidents. Classify emergencies according to their type and severity. Prioritize simultaneous emergencies. Assign suitable available response teams. Represent the city using a graph-based road network. Find efficient routes between teams and emergency locations. Manage unavailable teams and pending emergencies. Maintain team and resource status. Generate a summary of emergency response operations

Milestones

Requirement Analysis – Identify emergency types, teams, resources, locations, and system operations. System Design – Design classes, data structures, algorithms, and system architecture. Data Structure Implementation – Implement arrays, linked lists, stacks, queues, priority queues, trees, and graphs. Algorithm Implementation – Implement searching, sorting, BFS, DFS, and Dijkstra's algorithm. OOP Implementation – Implement classes, inheritance, polymorphism, encapsulation, and abstraction. Integration – Combine all modules into the complete simulator. Testing and Evaluation – Test different emergency scenarios and evaluate system behavior.

Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

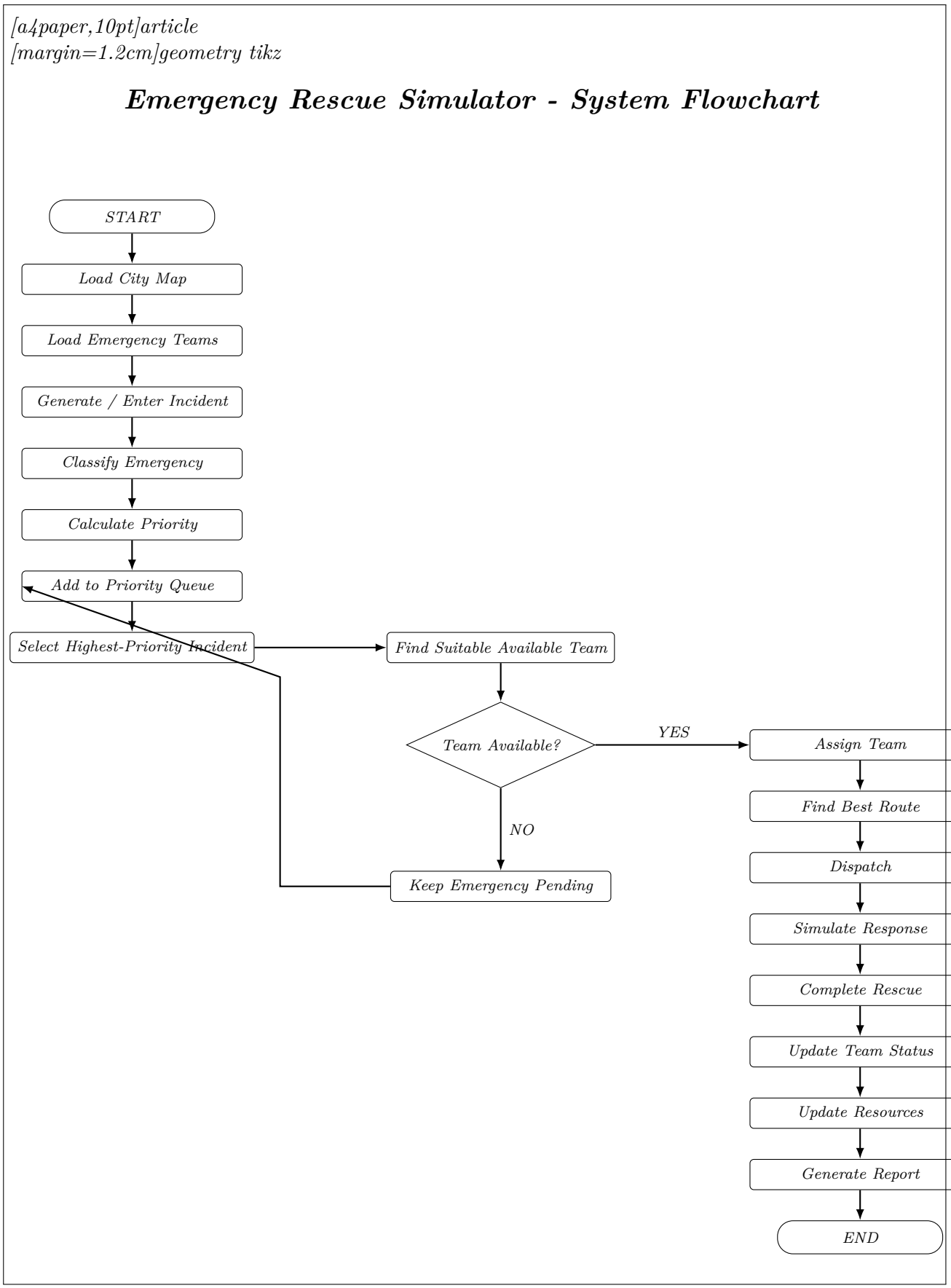
The project will be developed as a C/C++ based console application. C will be used to implement and demonstrate fundamental Data Structures and Algorithms, while C++ will be used to develop the object-oriented application layer.

The city will be represented as a weighted graph, where locations are vertices and roads are edges with associated distances. Emergency incidents will be stored and managed using appropriate data structures. A queue can represent incoming emergency requests, while a priority queue will prioritize incidents according to severity. Linked lists can maintain dynamically changing records, and stacks can maintain operation or route history.

The C++ implementation will follow OOP principles. Classes will represent entities such as Emergency, EmergencyTeam, FireTeam, MedicalTeam, PoliceTeam, RescueTeam, Location, and ResourceManager. Inheritance, polymorphism, encapsulation, abstraction, constructors, and function overloading will be incorporated where appropriate.

System Architecture (High Level Diagram)(2 pts)

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Project Outcome / Deliverables (1 pts)

The primary outcome of the project will be a functional Emergency Rescue Simulator capable of simulating the basic operations of an emergency response control center.
The project will deliver:
A C/C++ based menu-driven simulation application.
Route calculation using graph algorithms, particularly Dijkstra's algorithm.
Resource and team-status management. Response operation summaries and reports.

Assumptions

The simulator will operate using a predefined and simplified city map rather than real-time geographic data. Emergency locations, team locations, road distances, and available resources will initially be provided as input or predefined data. Each emergency is assumed to have a defined type and severity level. Teams are assumed to have predefined capabilities and availability status. Road distances are assumed to remain constant, and real-time traffic, weather, GPS, communication delays, and other unpredictable factors are not considered.

References

Reema Thareja, Data Structures Using C, Oxford University Press. — Used for understanding arrays, linked lists, stacks, queues, trees, graphs, and related operations.
GeeksforGeeks — Data Structures and Algorithms